

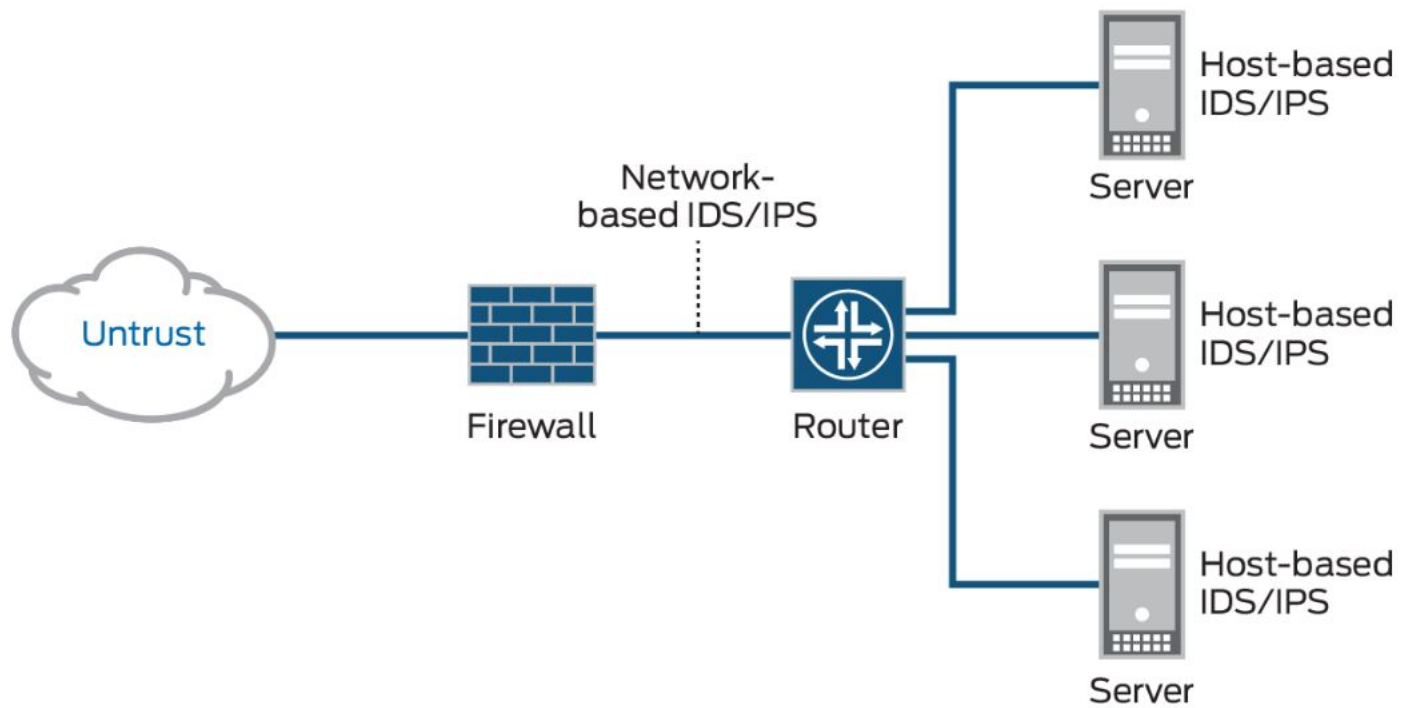
Intrusion Detection System

- Intrusion detection is the process of monitoring the events occurring in your network and analyzing them for signs of possible incidents, violations, or imminent threats to your security policies. Intrusion prevention is the process of performing intrusion detection and then stopping the detected incidents. These security measures are available as intrusion detection systems (IDS) and intrusion prevention systems (IPS), which become part of your network to detect and stop potential incidents.
- Intrusion: Attempting to break into or misuse your system.
- Intrusion can be a physical, system or remote intrusion.

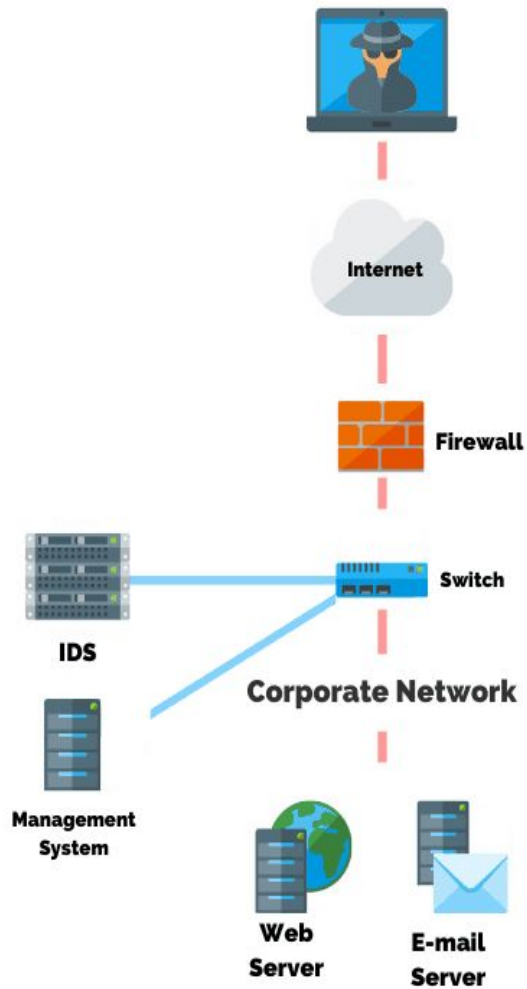
- An IDS is either a hardware device or software application that uses known intrusion signatures to detect and analyze both inbound and outbound network traffic for abnormal activities.

This is done through:

- System file comparisons against malware signatures
- Scanning processes that detect signs of harmful patterns
- Monitoring user behavior to detect malicious intent
- Monitoring system settings and configurations

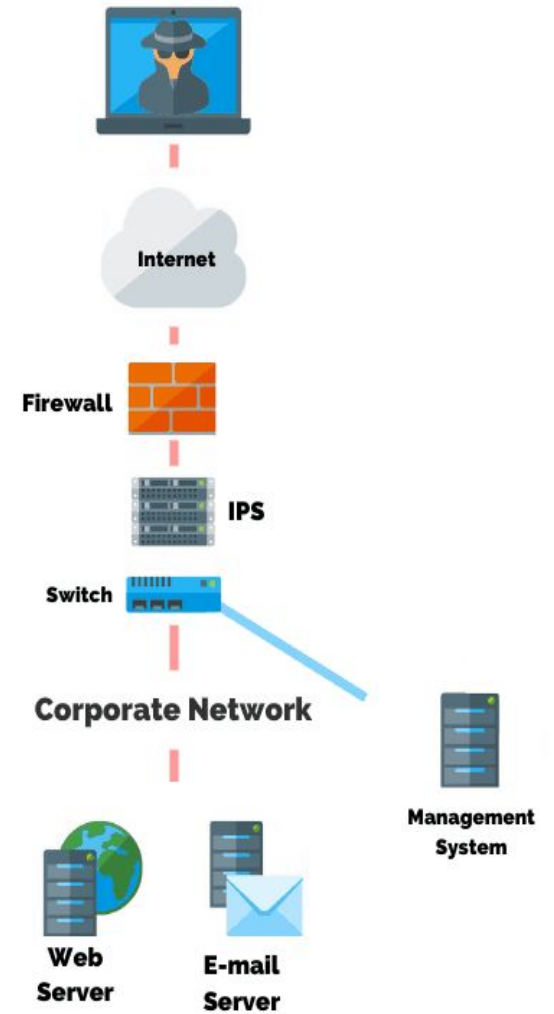


Intrusion Detection System (IDS)



VS

Intrusion Prevention System (IPS)



How Does IDS Work?

□ Intrusion Detection Systems(IDS)

- There are different ways of classifying IDS
 - Anomaly detection
 - Signature based misuse
 - Host based
 - Network based
 - Stack based
- Intrusion Detection Systems look for attack signatures, which are specific patterns that usually indicate malicious or suspicious intent.

Anomaly based IDS

- This IDS models the normal usage of the network as a noise characterization
- Anything distinct from the noise is assumed to be an intrusion activity.
 - e.g. flooding a host with lots of packet.
- The primary strength is its ability to recognize novel attacks.
- These generate many false alarms and hence compromise the effectiveness and this will be consider as a weak of the IDS,

Signature based IDS

- This IDS process an attacked description that can be matched to sensed attack
- The question of what information is relevant to an IDS depends upon what it is trying to detect.
 - e.g. DNS, FTP etc..
- ID system is programmed to interpret a certain series of packets, or a certain piece of data contained in those packets as an attack.
 - e.g. IDS watches web servers might be programmed to look for the string that would be a program attack.

Drawbacks of signature based IDS

- Have to programmed again for every new pattern to be detected.
- Suffer from false alrms

Host/Application based IDS

- The host operating system or the application logs in the audit information
- These audit information includes events like the use of identification and authentication mechanisms, file opens and program executions, admin activities etc.

□ **Some Drawbacks**

- The kind of information needed to be logged in is a matter of experience.
- Unselective logging of messages may greatly increase the audit and analysis burdens.

Stack based IDS

- They are integrated closely with the TCP/IP stack, allowing packets to be watched as they traverse their way up the OSI layers.
- This allows the IDS to pull the packets from the stack before the OS or the application have a chance to process the packets.

Network based IDS

- This IDS looks for attack signatures in network traffic via a promiscuous interface.
- A filter is usually applied to determine which traffic will be discarded or passed on to an attack recognition module. This helps to filter out known un-malicious traffic.